Web Appendix

Algorithm-assisted personalized risk communication to encourage flu vaccination in the USA: three randomized field trials

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1 Web appendix notes

This section includes additional analyses of the primary vaccination outcome for each study that *were not referenced* in the main text. For additional analyses that were referenced in the main text, see the Supplementary Information (SI).

All preregistered analyses of the primary outcome are included here or in the SI. Note that preregistered analyses differed across studies.

2 Study 1 additional analyses

2.1 Race and ethnicity

2.1.1 Primary outcome by study arm, race, and ethnicity

Web Appendix Table 1. Vaccination rates by study arm and race for Study 1.

Race	Passive control	High risk only	High risk medical records	High risk algorithm
American Indian or Alaska Native	0.300 n = 20	0.450 n = 20	0.500 n = 12	0.389 n = 18
Asian	0.704 n = 27	0.500 n = 28	0.652 n = 23	0.591 n = 22
Native Hawaiian or Other Pacific Islander	0.312 n = 16	0.391 n = 23	0.438 n = 16	0.304 n = 23
Black or African American	0.342 n = 336	0.386 n = 347	0.403 n = 355	0.382 n = 372
White	0.515 n = 9233	0.530 n = 9192	0.533 n = 9229	0.531 n = 9249
Unknown	0.625 n = 8	0.417 n = 12	0 n = 0	0.636 n = 11

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and racial group.

Web Appendix Table 2. Vaccination rates by study arm and ethnicity for Study 1.

Ethnicity	Passive control	High risk only	High risk medical records	High risk algorithm
Hispanic or Latino	0.348 n = 339	0.426 n = 406	0.384 n = 354	0.404 n = 362
Not Hispanic or Latino	0.514 n = 9289	0.528 n = 9196	0.533 n = 9272	0.530 n = 9314
Unknown	0.417 n = 12	0.450 n = 20	0.533 n = 15	0.326 n = 19

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and ethnic group.

2.1.2 Primary outcome regression with race/ethnicity as moderator

We tested for differences in the primary outcome as a function of risk messaging and race/ethnicity by running an OLS regression with a combined race/ethnicity predictor variable set to 1 for patients who were both white and non-Hispanic and 0 for patients who were not both white and non-Hispanic (or for whom race or ethnicity were unknown).

While white non-Hispanic patients (n = 35,602) were 16 percentage points more likely to get vaccinated than patients who were not white and non-Hispanic (n = 2,996) in the passive control arm, there was not a significant interaction between risk message and race/ethnicity (Web Appendix Table 3). If anything, there was a non-significant trend toward lower risk message effectiveness among white non-Hispanic patients.

Web Appendix Table 3. Study 1 OLS regression results for vaccination by study arm and race/ethnicity.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.360	0.323 – 0.397	19.235	38594	< 0.001
High risk messages	0.050	0.008 – 0.092	2.344	38594	.019
Race/ethnicity	0.160	0.122 – 0.198	8.224	38594	< 0.001
High risk messages * race/ethnicity	-0.035	-0.079 – 0.009	-1.569	38594	.117

Note. High risk messages is a binary variable coded as 1 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) and 0 for patients in the passive control arm. Race/ethnicity is a binary variable coded as 1 for patients who were both white and non-Hispanic and 0 for those who were not white and non-Hispanic, for whom race/ethnicity were unknown.

2.2 Sex

2.2.1 Primary outcome by study arm and sex

Web Appendix Table 4. Vaccination rates by study arm and sex for Study 1.

Sex	Passive control	High risk only	High risk medical records	High risk algorithm
Female	0.497 n = 6278	0.516 n = 6285	0.513 n = 6312	0.510 n = 6266
Male	0.530 n = 3362	0.539 n = 3337	0.556 n = 3329	0.552 n = 3429

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and sex.

2.2.2 Primary outcome regression with sex as moderator

We ran an OLS regression to test whether the primary outcome differed by study arm (risk message vs. no risk message) and sex. While males were 3.4 percentage points more

likely to get vaccinated than females in the passive control arm, there was not a significant interaction between risk messages and study arm (Web Appendix Table 5).

Web Appendix Table 5. Study 1 OLS regression results for vaccination by study arm and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.497	0.484 – 0.509	78.825	38594	<0.001
High risk message	0.016	0.002 – 0.031	2.249	38594	0.025
Sex	0.034	0.013 – 0.055	3.157	38594	0.002
High risk message * Sex	0.002	-0.022 – 0.026	0.186	38594	0.853

Note. High risk message is a binary variable coded as 1 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) and 0 for patients in the Passive Control arm. Sex is a binary variable coded as 1 for males and 0 for females.

2.3 Age

2.3.1 Age 65+ vs. under 65

While older patients tend to be aware of their increased vulnerability, younger patients may be more surprised to learn of their high-risk status. In a preregistered exploratory analysis, we examined the differential response of patients above vs. below age 65 by introducing this age variable as moderator and assessing if it interacts with experimental group to affect vaccination behavior.

Regression results are shown in Web Appendix Table 6. Although patients ages 65+ (n = 14,277) were 23 percentage points more likely to get vaccinated than patients under age 65 (n = 24,321) in the passive control arm, risk messages were not differentially effective in patients ages 65+ compared with patients under age 65.

Web Appendix Table 6. Study 1 OLS regression results for vaccination by study arm and age group (65+ vs. under 65).

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.423	0.411 – 0.435	67.836	38594	< .001
High risk message	0.014	0.000 – 0.028	1.978	38594	.048
Age group	0.231	0.211 – 0.251	22.504	38594	< .001
High risk message * Age group	0.007	-0.016 – 0.031	0.628	38594	.530

Note. High risk message is a binary variable coded as 1 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) and 0 for patients in the passive control arm. Age group is a binary variable coded as 1 for patients ages 65+ and 0 for patients under age 65.

2.3.2 Binned age

Our preregistrations for Studies 1 and 2 included an analysis of continuous age. Prior to studies 3 and 4, we decided it would be more appropriate to use binned age (18–24, 25–34, 35–

44,45–54, 65+) rather than continuous age, due to possible nonlinear relationships with age. Therefore, we preregistered binned age analyses for Studies 3 and 4 and report binned age rather than continuous age analyses for Studies 1 and 2.

Sample sizes for each age bin were as follows: 18–24 *n* = 2,505, 25–34 *n* = 4,629, 35–44 *n* = 4,629, 45–54 *n* = 5,656, 55–64 *n* = 7,409, 65+ *n* = 14,277.

We ran an OLS regression to test whether risk messages were differentially effective as a function of age group. Age bin did not interact significantly with message condition in Study 1 (Web Appendix Table 7).

Web Appendix Table 7. Study 1 OLS regression results for vaccination by study arm and binned age.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.260	0.222 – 0.298	13.379	38586	<0.001
High risk messages	0.034	-0.009 – 0.078	1.536	38586	0.125
Age 25-34	0.042	-0.006 – 0.090	1.697	38586	0.090
Age 35-44	0.105	0.058 – 0.152	4.384	38586	<0.001
Age 45-54	0.184	0.138 – 0.230	7.913	38586	<0.001
Age 55-64	0.302	0.258 – 0.346	13.530	38586	<0.001
Age 65+	0.394	0.353 – 0.435	18.755	38586	<0.001
High risk messages*Age 25-34	-0.022	-0.077 – 0.033	-0.776	38586	0.438
High risk messages*Age 35-44	-0.024	-0.078 – 0.031	-0.854	38586	0.393
High risk messages*Age 45-54	-0.012	-0.065 – 0.040	-0.457	38586	0.648
High risk messages*Age 55-64	-0.024	-0.074 – 0.027	-0.929	38586	0.353
High risk messages*Age 65+	-0.013	-0.060 – 0.035	-0.521	38586	0.602

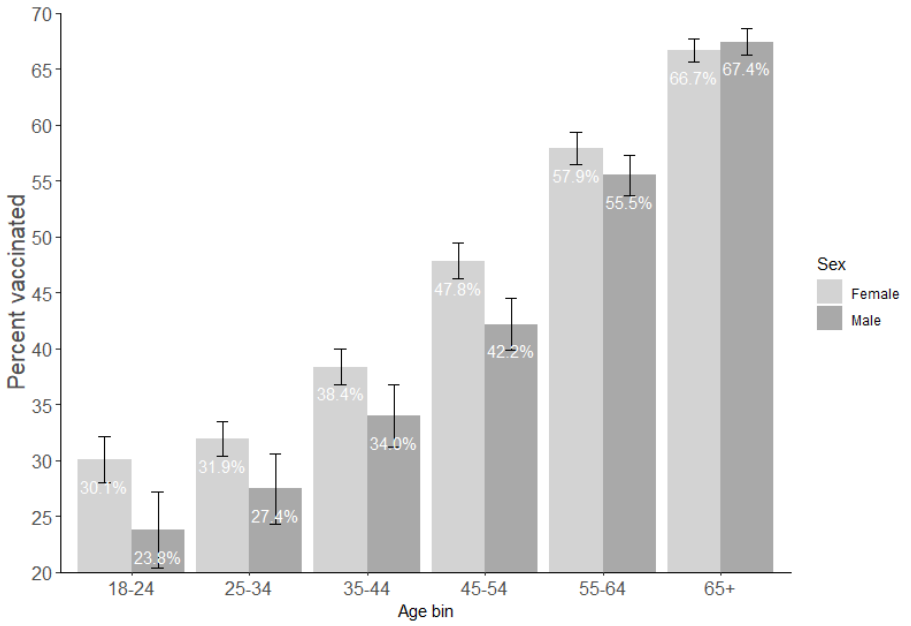
Note. High risk messages is a binary variable coded as 1 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) and 0 for patients in the passive control arm. The reference age group is age 18-24.

2.4 Age and Sex

We tested for effects of study arm with sex, age bins, and interactions between sex and age as covariates in a preregistered exploratory analysis.

With the age, sex, and interaction covariates included in the regression, there was a significant intervention effect of high-risk messages compared with passive control messages. The effect was stronger than the effect reported in the main text without covariates (Web Appendix Table 8).

We also found significant interactive effects between age group and sex. In those age 65+, the difference in vaccination between males and females was significantly higher than that in the 18–24 age bin. In general, for younger age bins, females appear to be more likely to get vaccinated than males. However, males, relative to females, become increasingly more likely to be vaccinated with increasing age (Web Appendix Fig. 1, Web Appendix Table 8).



Web Appendix Fig. 1. Vaccination rate as a function of sex and age bin in Study 1.

The percentage of male and female patients in each age bin who were vaccinated during the primary outcome time frame. Error bars indicate 95% confidence intervals.

Web Appendix Table 8. Study 1 OLS regression results for vaccination by study arm, age bin, and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.287	0.264 – 0.310	24.442	38585	<0.001
High risk messages	0.018	0.007 – 0.029	3.247	38585	0.001
Age 25-34	0.019	-0.008 – 0.046	1.361	38585	0.174
Age 35-44	0.083	0.056 – 0.110	6.102	38585	<0.001
Age 45-54	0.178	0.151 – 0.204	13.277	38585	<0.001
Age 55-64	0.279	0.253 – 0.304	21.345	38585	<0.001
Age 65+	0.366	0.342 – 0.390	30.073	38585	<0.001
Sex	-0.063	-0.107 – -0.019	-2.811	38585	0.005
Age 25-34*Sex	0.018	-0.039 – 0.076	0.618	38585	0.536
Age 35-44*Sex	0.020	-0.035 – 0.074	0.702	38585	0.483
Age 45-54*Sex	0.006	-0.045 – 0.058	0.246	38585	0.805
Age 55-64*Sex	0.039	-0.011 – 0.088	1.540	38585	0.123
Age 65+*Sex	0.071	0.024 – 0.118	2.958	38585	0.003

Note. High risk messages is a binary variable coded as 1 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) and 0 for patients in the passive control arm. The reference age group is age 18-24. Sex is a binary variable coded as 1 for males and 0 for females.

2.5 Modalities

2.5.1 Primary outcome removing patients who were not eligible for any messages

One of our study exclusion criteria was “Has opted out of receiving communications from Geisinger via all of the modalities being tested.” In Study 1, we performed this exclusion prior to randomization. However, we received updated lists of patients who were eligible for portal and SMS messages during the study, closer to when messages were sent in those modalities. There were 37 patients who were not eligible for communications in any modalities as of the most updated data pulls for each modality.

We removed these 37 patients and re-ran the primary analysis reported in the main text contrasting vaccination rates in patients who received high-risk messages with those in the passive control group. The result did not meaningfully change: patients who received risk messages were 1.8 percentage points more likely to be vaccinated (OLS regression, b = .018 [95% CI: 0.006, 0.029], t(38559) = 2.99, p = .003).

2.5.2 Primary outcome as a function of modality eligibility

Patients who received an intervention message may have received communications via up to 3 modalities. In a preregistered analysis, we ran an OLS regression to evaluate the impact

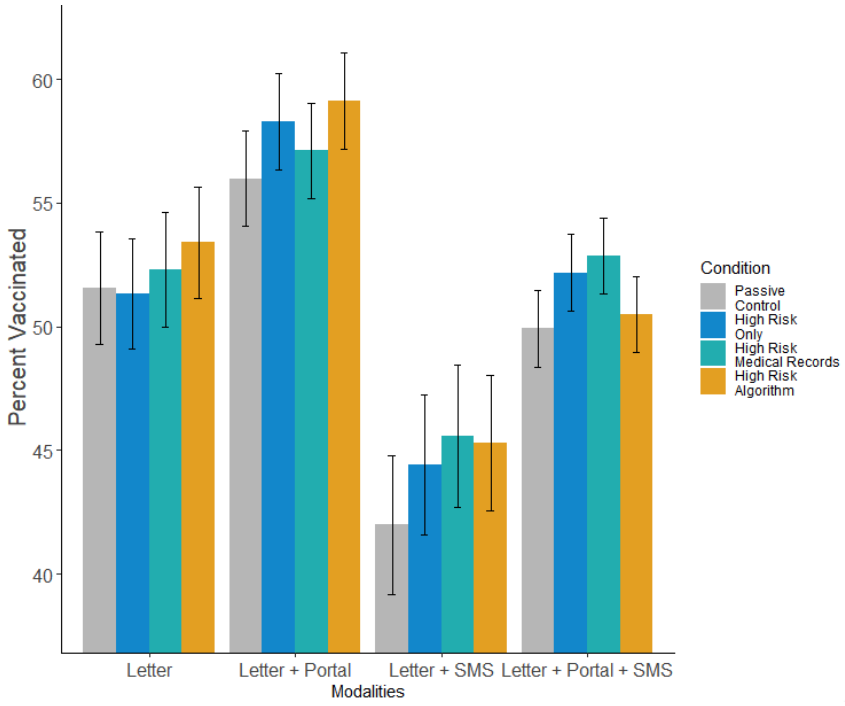
of modality on the primary vaccination outcome, assessing pairwise comparisons of the following modality combinations: letter only, letter + portal, letter + SMS, or letter + portal + SMS. We excluded from analysis patients who did not experience one of these modality combinations and patients in the passive control group.

There were significant differences in vaccination across modality combinations (Web Appendix Table 9). However, modalities were not randomly assigned so underlying differences across groups eligible for different modalities likely contribute to the different vaccination rates observed. Consistent with this idea, vaccination rates in the passive control arm were similar to those in active arms within each modality group (Web Appendix Fig. 2).

Web Appendix Table 9. Pairwise comparisons of vaccination rates across modality groups in Study 1.

Contrast	Estimate	95% CI	*t*	*df*	*p*
Letter - (Letter + portal)	-0.058	-0.076 – -0.041	-6.633	28763	< .001
Letter - (Letter + SMS)	0.072	0.052 – 0.093	6.811	28763	< .001
Letter - (Letter + portal + SMS)	0.005	-0.011 – 0.021	0.623	28763	0.533
(Letter + portal) - (Letter + SMS)	0.131	0.111 – 0.151	12.937	28763	< .001
(Letter + portal) - (Letter + portal + SMS)	0.063	0.049 – 0.078	8.644	28763	< .001
(Letter + SMS) - (Letter + portal + SMS)	-0.067	-0.086 – -0.049	-7.131	28763	< .001

Note. The reference group is listed first for each comparison (e.g., in the first row, letter is coded as 0 and letter + portal is coded as 1).



Web Appendix Fig. 2. Vaccination rate as a function of modality and study arm in Study 1.

The primary outcome plotted as a function of study arm and the modalities for which patients were eligible. Error bars indicate 95% confidence intervals.

3 Study 2 additional analyses

3.1 Race and ethnicity

3.1.1 Primary outcome by study arm, race, and ethnicity

Web Appendix Table 10. Vaccination rates by study arm and race for Study 2.

Race	Passive control	Active control	High risk only	High risk medical records + reasons	High risk algorithm + reasons
American Indian or Alaska Native	0.182 n = 11	0.188 n = 16	0.100 n = 10	0.125 n = 16	0.467 n = 15
Asian	0.308 n = 26	0.278 n = 18	0.318 n = 22	0.500 n = 24	0.269 n = 26
Native Hawaiian or Other Pacific Islander	0.286 n = 14	0.083 n = 12	0.100 n = 10	0.400 n = 15	0.176 n = 17
Black or African American	0.199 n = 337	0.184 n = 277	0.239 n = 284	0.193 n = 316	0.169 n = 290
White	0.321 n = 8058	0.335 n = 8185	0.350 n = 8158	0.346 n = 8086	0.343 n = 8163
Unknown	0.050 n = 20	0.357 n = 14	0.286 n = 14	0.385 n = 13	0.071 n = 14

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. *N* reflects the number of patients in the indicated arm and racial group.

Web Appendix Table 11. Vaccination rates by study arm and ethnicity for Study 2.

Ethnicity	Passive control	Active control	High risk only	High risk medical records + reasons	High risk algorithm + reasons
Hispanic or Latino	0.210 n = 362	0.227 n = 335	0.192 n = 355	0.232 n = 336	0.205 n = 347
Not Hispanic or Latino	0.319 n = 8076	0.334 n = 8160	0.353 n = 8116	0.346 n = 8113	0.341 n = 8156
Unknown	0.500 n = 28	0.333 n = 27	0.185 n = 27	0.238 n = 21	0.364 n = 22

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. N reflects the number of patients in the indicated arm and ethnic group.

3.1.2 Primary outcome regression with race/ethnicity as moderator

As with Study 1, we tested for differences in the primary outcome as a function of risk messaging and race/ethnicity. Again, the combined race/ethnicity predictor variable was set to 1 for patients who were both white and non-Hispanic and 0 for patients who were not both white

and non-Hispanic (or for whom race or ethnicity were unknown). Here, we performed one regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms).

Results are reported in Web Appendix Table 12. White and non-Hispanic patients (n = 39,049) were 13.8 percentage points more likely to get vaccinated than those who were not white and non-Hispanic (n = 3,432) in the high-risk message arms. There was not a significant interaction between study arm and race/ethnicity.

Web Appendix Table 12. Study 2 OLS regression results for vaccination by study arm and race/ethnicity.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.214	0.194 – 0.234	20.570	42475	<0.001
Study arm: Passive control vs. high-risk messages	0.001	-0.039 – 0.040	0.048	42475	0.962
Study arm: Active control vs. high-risk messages	0.000	-0.041 – 0.042	0.013	42475	0.990
Race/ethnicity	0.138	0.117 – 0.159	12.724	42475	<0.001
Study arm: Passive control vs. high risk * Race/ethnicity	-0.028	-0.069 – 0.013	-1.331	42475	0.183
Study arm: Active control vs. high risk * Race/ethnicity	-0.013	-0.056 – 0.030	-0.601	42475	0.548

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Race/ethnicity is a binary variable coded as 1 for patients who were both white and non-Hispanic and 0 for those who were not white and non-Hispanic, for whom race/ethnicity were unknown.

3.2 Sex

3.2.1 Primary outcome by study arm and sex

Web Appendix Table 13. Vaccination rates by study arm and sex for Study 2.

Sex	Passive control	Active control	High risk only	High risk medical records + reasons	High risk algorithm + reasons
Female	0.307 n = 5670	0.318 n = 5794	0.336 n = 5746	0.327 n = 5740	0.332 n = 5816
Male	0.332 n = 2796	0.354 n = 2728	0.365 n = 2752	0.370 n = 2730	0.343 n = 2709

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. N reflects the number of patients in the indicated arm and sex.

3.2.2 Primary outcome regression with sex as moderator

We tested whether males and females differently responded to risk messages in an OLS regression. We ran one regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms).

Results are reported in Web Appendix Table 14. Males were 2.7 percentage points more likely to get vaccinated than females in the high-risk message arms. There was not a significant interaction between study arm and sex.

Web Appendix Table 14. Study 2 OLS regression results for vaccination by study arm and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.332	0.325 – 0.339	92.684	42475	<0.001
Study arm: Passive control vs. high-risk messages	-0.025	-0.039 – -0.011	-3.482	42475	<0.001
Study arm: Active control vs. high-risk messages	-0.014	-0.028 – -0.000	-2.016	42475	0.044
Sex	0.027	0.015 – 0.040	4.341	42475	<0.001
Study arm: Passive control vs. high risk * Sex	-0.002	-0.027 – 0.022	-0.191	42475	0.848
Study arm: Active control vs. high risk * Sex	0.009	-0.015 – 0.034	0.749	42475	0.454

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Sex is a binary variable coded as 1 for males and 0 for females.

3.3 Age

3.3.1 Age 65+ vs. under 65

While older patients tend to be aware of their increased vulnerability, younger patients may be more surprised to learn of their high-risk status. In a preregistered exploratory analysis, we examined the differential response of patients above vs. below age 65 by introducing this age variable as a moderator and assessing if it interacts with experimental group to affect vaccination behavior. We ran one OLS regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms). We crossed this study arm variable with age group (65+ vs. under 65).

Regression results are shown in Web Appendix Table 15. Although patients ages 65+ (n = 13,782) were 29.2 percentage points more likely to get vaccinated than patients under age 65

(n = 28,699) in the high-risk message arms, risk messages were not differentially effective in patients ages 65+ compared with patients under age 65.

Web Appendix Table 15. Study 2 OLS regression results for vaccination by study arm and age group (65+ vs. under 65).

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.246	0.240 – 0.253	71.714	42475	<0.001
Study arm: Passive control vs. high-risk messages	-0.023	-0.037 – -0.010	-3.414	42475	0.001
Study arm: Active control vs. high-risk messages	-0.013	-0.026 – 0.001	-1.817	42475	0.069
Age 65+	0.292	0.280 – 0.304	48.266	42475	<0.001
Study arm: Passive control vs. high-risk messages * Age 65+	-0.006	-0.030 – 0.018	-0.485	42475	0.628
Study arm: Active control vs. high-risk messages * Age 65+	-0.002	-0.026 – 0.022	-0.161	42475	0.872

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Age group is a binary variable coded as 1 for patients ages 65+ and 0 for patients under age 65.

3.3.2 Binned age

As in Study 1, we examined whether response to messages differed as a function of age group. Sample sizes for each age bin were as follows: 18–24 *n* = 3,377, 25–34 *n* = 5,718, 35–44 *n* = 5,937, 45–54 *n* = 6,218, 55–64 *n* = 7,449, 65+ *n* = 13,782.

To test whether age bin moderated study arm effects, we ran one OLS regression testing for an interaction between age bin and a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms). For the age bin variable, the Age 18–24 bin was the reference group.

Results are reported in Web Appendix Table 16. Vaccination rates increase with increasing age. Although the passive control vs. high-risk messages X Age 55-64 bin technically surpassed the $p < .05$ threshold (indicating the difference between passive control and high-risk messages was different in this age bin compared with the age 18–24 age bin), there were no clear overall interaction patterns.

Web Appendix Table 16. Study 2 OLS regression results for vaccination by study arm and binned age.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.137	0.118 – 0.157	13.971	42463	<0.001
Study arm: Passive control vs. high-risk messages	-0.007	-0.046 – 0.033	-0.331	42463	0.741
Study arm: Active control vs. high-risk messages	-0.038	-0.077 – 0.001	-1.915	42463	0.056
Age 25-34	0.022	-0.002 – 0.047	1.784	42463	0.075
Age 35-44	0.063	0.039 – 0.087	5.100	42463	<0.001
Age 45-54	0.126	0.102 – 0.150	10.291	42463	<0.001
Age 55-64	0.248	0.225 – 0.271	20.909	42463	<0.001
Age 65+	0.401	0.379 – 0.422	36.497	42463	<0.001
Study arm: Passive control vs. high-risk messages * Age 25-34	-0.020	-0.069 – 0.029	-0.804	42463	0.422
Study arm: Active control vs. high-risk messages * Age 25-34	0.034	-0.015 – 0.083	1.372	42463	0.170
Study arm: Passive control vs. high-risk messages * Age 35-44	0.006	-0.043 – 0.055	0.229	42463	0.819
Study arm: Active control vs. high-risk messages * Age 35-44	0.017	-0.031 – 0.066	0.695	42463	0.487
Study arm: Passive control vs. high-risk messages * Age 45-54	-0.008	-0.056 – 0.041	-0.313	42463	0.754
Study arm: Active control vs. high-risk messages * Age 45-54	0.047	-0.001 – 0.096	1.922	42463	0.055
Study arm: Passive control vs. high-risk messages * Age 55-64	-0.049	-0.096 – -0.002	-2.060	42463	0.039
Study arm: Active control vs. high-risk messages * Age 55-64	0.021	-0.026 – 0.068	0.885	42463	0.376
Study arm: Passive control vs. high-risk messages * Age 65+	-0.023	-0.066 – 0.021	-1.023	42463	0.306
Study arm: Active control vs. high-risk messages * Age 65+	0.024	-0.020 – 0.067	1.066	42463	0.286

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. The reference age group is age 18-24.

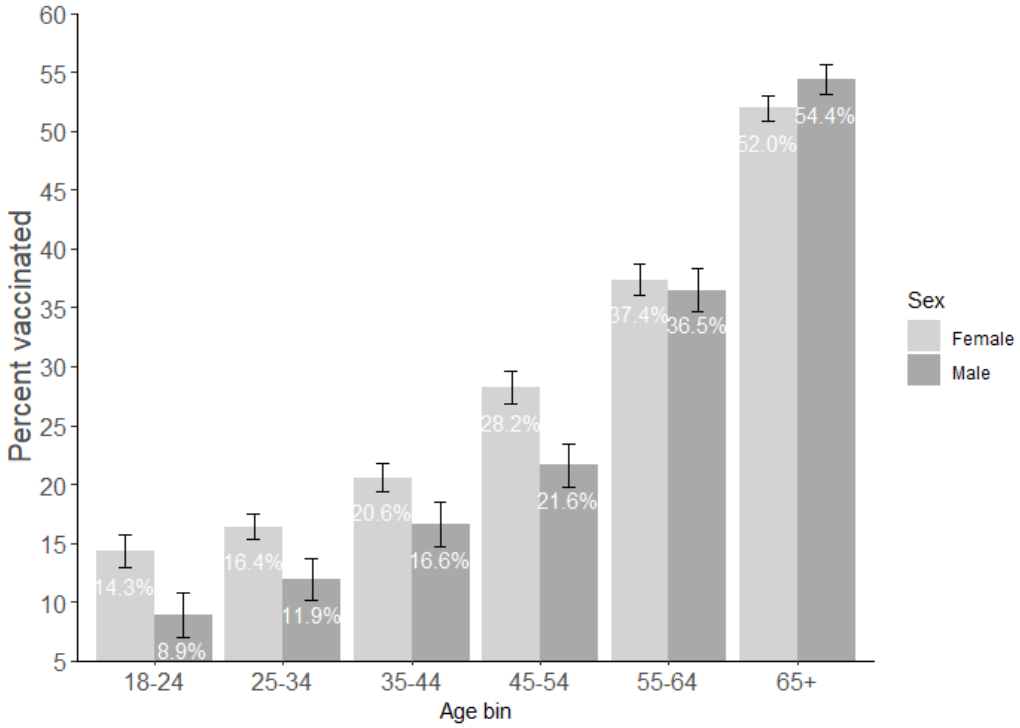
3.4 Age and sex

In a preregistered robustness check, we tested for effects of study arm with sex, age bins, and interactions between sex and age as covariates. As in the above exploratory analyses of Study 2 data, we ran one OLS regression testing for an interaction between age bin and a

categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. We crossed this variable with age bin.

With the age, sex, and interaction covariates included in the regression, there was a significant intervention effect of high-risk messages compared with both active and passive control messages. Both effects were stronger than the effects reported in the main text without covariates (Web Appendix Table 17).

Similar to the pattern in Study 1, we found significant interactive effects between age group and sex, with a significantly different pattern of vaccination by sex in the 55–64 and 65+ age bins compared with the 18–24 age bin. Generally, females appear to be more likely to be vaccinated at younger ages, while males are to be increasingly likely to be vaccinated (relative to females) with increasing age (Web Appendix Fig. 3, Web Appendix Table 17).



Web Appendix Fig. 3. Vaccination rate as a function of sex and age bin in Study 2.

The percentage of male and female patients in each age bin who were vaccinated during the primary outcome time frame. Error bars indicate 95% confidence intervals.

Web Appendix Table 17. Study 2 OLS regression results for vaccination by study arm, age bin, and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.150	0.132 – 0.168	16.506	42467	<0.001
Study arm: Passive control vs. high-risk messages	-0.026	-0.036 – -0.015	-4.567	42467	<0.001
Study arm: Active control vs. high-risk messages	-0.013	-0.024 – -0.002	-2.267	42467	0.023
Age 25-34	0.021	-0.001 – 0.043	1.888	42467	0.059
Age 35-44	0.063	0.041 – 0.085	5.657	42467	<0.001
Age 45-54	0.140	0.118 – 0.162	12.455	42467	<0.001
Age 55-64	0.231	0.210 – 0.253	21.007	42467	<0.001
Age 65+	0.377	0.357 – 0.397	36.961	42467	<0.001
Sex	-0.054	-0.087 – -0.020	-3.092	42467	0.002
Age 25-34 * Sex	0.009	-0.034 – 0.053	0.414	42467	0.679
Age 35-44 * Sex	0.014	-0.029 – 0.057	0.630	42467	0.529
Age 45-54 * Sex	-0.013	-0.054 – 0.029	-0.597	42467	0.551
Age 55-64 * Sex	0.045	0.005 – 0.085	2.188	42467	0.029
Age 65+ * Sex	0.078	0.041 – 0.115	4.121	42467	<0.001

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. The reference age group is age 18–24. Sex is a binary variable coded as 1 for males and 0 for females.

3.5 Modalities

3.5.1 Primary outcome removing patients who were not eligible for any messages

As in Study 1, we applied the exclusion criterion, “Has opted out of receiving communications from Geisinger via all of the modalities being tested” prior to randomization but received updated lists of patients who could not be contacted via portal or SMS. There were 126 patients who were included in analyses in the main text but could not be contacted via any study modality. Additionally, we

We reran the primary outcome analyses reported in the main text, but without these patients who could not be contacted. Consistent with the effect reported in the main text, patients who were sent risk messages were 2.6 percentage points more likely to be vaccinated than patients in the passive control group (OLS regression: b = .026, [95% CI: 0.015, 0.038], $t(33850) = 4.416$, $p < .001$). Additionally, those who were sent high-risk messages were 1.2 percentage points more likely to be vaccinated than patients in the active control group (OLS regression: b = .012, [95% CI: 0.000, 0.023], $t(33899) = 1.963$, $p = .050$). Notably, although the effect size did not substantially change from the analysis reported in the main text (1.1 percentage points in the main text, 1.2 percentage points here), the effect size reported here

reached statistical significance when the few patients who could not be contacted were removed from analysis.

3.5.2 Primary outcome as a function of modality eligibility

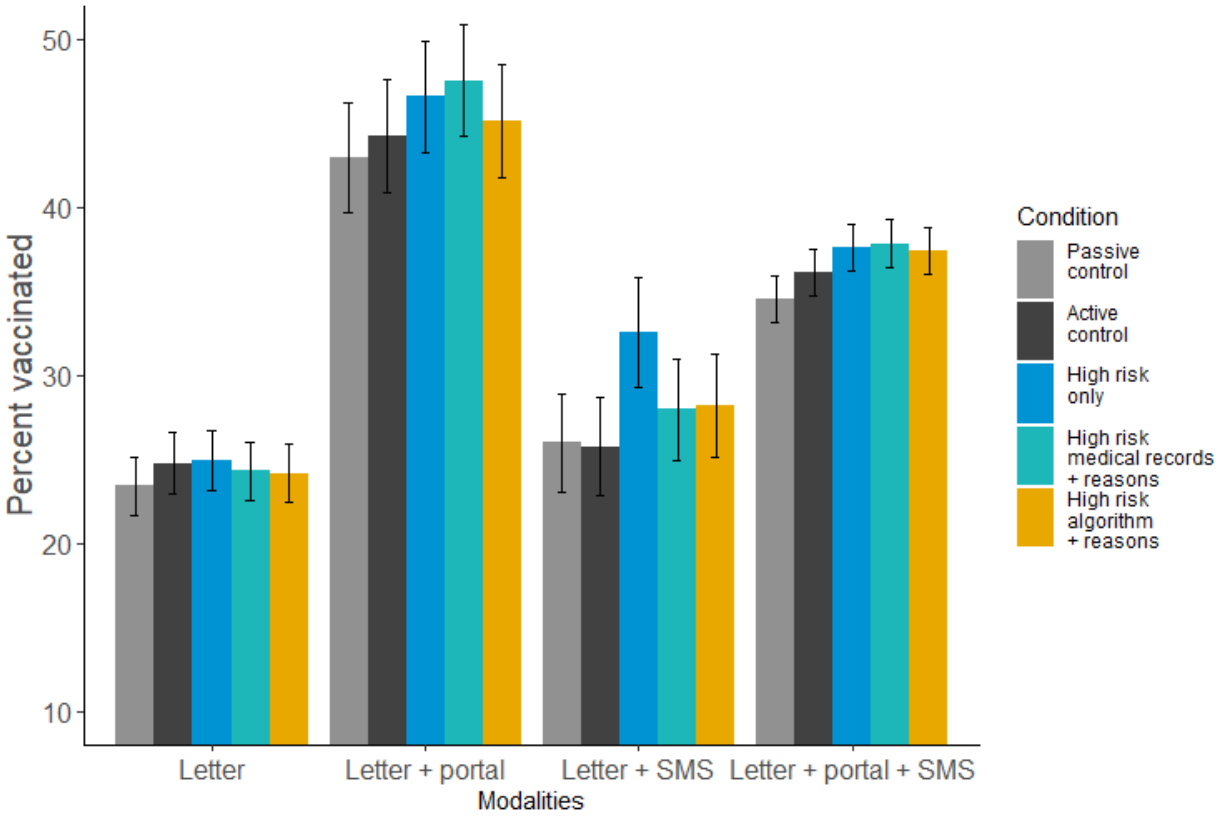
We ran a preregistered exploratory analysis to test whether overall vaccination varied as a function of the modalities patients were sent. As in Study 1, we assessed pairwise comparisons of vaccination in patients sent messages in the following modality combinations: letter only (*n* = 11,272), letter + portal (*n* = 4,268), letter + SMS (*n* = 4,159), or letter + portal + SMS (*n* = 22,654). Patients who did not experience one of these modality combinations and patients in the passive control group were not included in this analysis.

All pairwise comparisons across modality groups were statistically significant (Web Appendix Table 18), but modalities were not randomized and underlying differences in patient characteristics likely played a role in these differences. Consistent with this notion, vaccination rates in the passive control arm varied alongside those in active arms for all modality combinations (Web Appendix Fig. 4).

Web Appendix Table 18. Pairwise comparisons of vaccination rates across modality groups in Study 2.

contrast	Estimate	95% CI	*t*	*df*	*p*
Letter - (Letter + portal)	-0.214	-0.232–-0.195	-22.687	33895	<.001
Letter - (Letter + SMS)	-0.040	-0.059– -0.022	-4.232	33895	<.001
Letter - (Letter + portal + SMS)	-0.127	-0.139–-0.115	-21.047	33895	<.001
(Letter + portal) - (Letter + SMS)	0.173	0.151–0.196	15.165	33895	<.001
(Letter + portal) - (Letter + portal + SMS)	0.087	0.070–0.104	9.916	33895	<.001
(Letter + SMS) - (Letter + portal + SMS)	-0.087	-0.104–-0.069	-9.791	33895	<.001

N*ote.* The reference group is listed first for each comparison (e.g., in the first row, letter is coded as 0 and letter + portal is coded as 1).



Web Appendix Fig. 4. Vaccination rate as a function of modality and study arm in Study 2.

The primary outcome plotted as a function of study arm and the modalities for which patients were eligible. Error bars indicate 95% confidence intervals.

3.6 SMS timing

This preregistered exploratory analysis focused on the subgroup of patients who were sent SMS messages. As mentioned in the Methods section, due to delays outside our control, SMS messages were sent on 3 separate days - 15 days, 24 days, or 25 days after patient portal messages. We tested whether there was a difference in flu shot uptake or timing in those who received earlier messages (15 days following the portal message) compared with those who received later messages (24 or 25 days after the portal message).

To this end, we first ran an OLS regression testing for effects of SMS timing, message type, or their interaction, on flu shot uptake (Web Appendix Table 19). We did not find significant interactive effects of message timing on vaccination.

Web Appendix Table 19. Study 2 OLS regression results for vaccination by study arm and SMS timing.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.347	0.329 – 0.366	37.282	21310	<0.001
SMS timing	-0.003	-0.029 – 0.022	-0.244	21310	0.807
High risk only	0.024	-0.002 – 0.050	1.807	21310	0.071
High risk medical records	0.018	-0.008 – 0.044	1.336	21310	0.182
High risk algorithm	0.014	-0.012 – 0.040	1.027	21310	0.304
SMS timing * High risk only	-0.003	-0.039 – 0.034	-0.137	21310	0.891
SMS timing * High risk medical records	-0.002	-0.038 – 0.034	-0.106	21310	0.916
SMS timing * High risk algorithm	0.001	-0.035 – 0.037	0.054	21310	0.957

Note. SMS timing is a binary variable coded as 0 for an early message or 1 for a late message. Active control was the baseline group for message condition, so main effects or interactions with the high-risk message conditions are in comparison with the active control group. Sample sizes for message timing: early n = 10,332; late, n = 10,986. Sample sizes for message arms sent SMS messages and included in analysis: active control, n = 5,465; high risk only, n = 5,316; high risk medical records, n = 5,231; high risk algorithm, n = 5,306.

Second, we tested whether flu shot timing varied in patients who received early compared with late SMS messages. Among patients who were vaccinated and were sent SMS messages, there were no significant main effects of SMS timing on vaccination timing and there was not a significant interaction between SMS timing and condition on vaccination timing (Web Appendix Table 20).

Web Appendix Table 20. Study 2 OLS regression results for vaccination timing by study arm and SMS timing.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	25.999	25.014 – 26.984	51.734	7644	<0.001
SMS timing	-0.401	-1.777 – 0.975	-0.571	7644	0.568
High risk only	-0.468	-1.848 – 0.913	-0.664	7644	0.507
High risk medical records	0.662	-0.732 – 2.056	0.930	7644	0.352
High risk algorithm	0.152	-1.236 – 1.540	0.215	7644	0.830
SMS timing * High risk only	0.038	-1.891 – 1.967	0.039	7644	0.969
SMS timing * High risk medical records	0.346	-1.598 – 2.291	0.349	7644	0.727
SMS timing * High risk algorithm	0.040	-1.901 – 1.981	0.040	7644	0.968

Note. SMS timing is a binary variable coded as 0 for an early message or 1 for a late message. Active control was the baseline group for message condition, so main effects or interactions with the high-risk message conditions are in comparison to the active control group. Sample sizes for message timing groups among vaccinated patients: early n = 3,790; late n = 3,992; Sample sizes for message conditions among vaccinated patients who received SMS: active control, n = 1,889; high risk only, n = 1,958; high risk medical records, n = 1,896; high risk algorithm, n = 1,909.

4 Study 3 additional analyses

4.1 Race and ethnicity

4.1.1 Primary outcome by study arm, race, and ethnicity

Web Appendix Table 21. Vaccination rates by study arm and race for Study 3.

Race	Passive control	Active control	High risk only	High risk medical records	High risk algorithm
American Indian or Alaska Native	0.062 n = 16	0.118 n = 17	0.227 n = 22	0.154 n = 13	0.176 n = 17
Asian	0.412 n = 34	0.241 n = 29	0.343 n = 35	0.239 n = 46	0.286 n = 42
Native Hawaiian or Other Pacific Islander	0.150 n = 20	0.032 n = 31	0.108 n = 37	0.160 n = 25	0.188 n = 32
Black or African American	0.112 n = 304	0.144 n = 320	0.162 n = 302	0.132 n = 296	0.130 n = 322
White	0.243 n = 7397	0.265 n = 7379	0.286 n = 7374	0.280 n = 7376	0.272 n = 7373
More Than One Race	0 n = 1	0.500 n = 2	0 n = 1	0 n = 0	1.000 n = 1
Unknown	0.137 n = 51	0.242 n = 62	0.158 n = 38	0.317 n = 41	0.164 n = 73

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and racial group.

Web Appendix Table 22. Vaccination rates by study arm and ethnicity for Study 3.

Ethnicity	Passive control	Active control	High risk only	High risk medical records	High risk algorithm
Hispanic or Latino	0.165 n = 399	0.164 n = 391	0.168 n = 411	0.160 n = 400	0.176 n = 415
Not Hispanic or Latino	0.242 n = 7382	0.264 n = 7397	0.286 n = 7354	0.279 n = 7360	0.270 n = 7403
Unknown	0.214 n = 42	0.231 n = 52	0.364 n = 44	0.324 n = 37	0.262 n = 42

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and ethnic group.

4.1.2 Primary outcome regression with race/ethnicity as moderator

We again tested for differences in the primary outcome as a function of risk messaging and race/ethnicity. Like Studies 1 and 2, the combined race/ethnicity predictor variable was set

to 1 for patients who were both white and non-Hispanic and 0 for patients who were not both white and non-Hispanic (or for whom race or ethnicity were unknown). Here, we ran one regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms).

Results are reported in Web Appendix Table 23. White and non-Hispanic patients (n = 35,205) were 11.1 percentage points more likely to get vaccinated than those who were not white and non-Hispanic (n = 3,924) in the high-risk message arms. There was not a significant interaction between study arm and race/ethnicity.

Web Appendix Table 23. Study 3 OLS regression results for vaccination by study arm and race/ethnicity.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.173	0.156 – 0.191	19.252	39123	<0.001
Study arm: Passive control vs. high-risk messages	-0.020	-0.056 – 0.016	-1.106	39123	0.269
Study arm: Active control vs. high-risk messages	-0.001	-0.036 – 0.035	-0.051	39123	0.960
Race/ethnicity	0.111	0.092 – 0.129	11.662	39123	<0.001
Study arm: Passive control vs. high risk * Race/ethnicity	-0.017	-0.055 – 0.020	-0.898	39123	0.369
Study arm: Active control vs. high risk * Race/ethnicity	-0.015	-0.052 – 0.023	-0.768	39123	0.443

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Race/ethnicity is a binary variable coded as 1 for patients who were both white and non-Hispanic and 0 for those who were not white and non-Hispanic, for whom race/ethnicity were unknown.

4.2 Sex

4.2.1 Primary outcome by study arm and sex

Web Appendix Table 24. Vaccination rates by study arm and sex for Study 3.

Sex	Passive control	Active control	High risk only	High risk medical records	High risk algorithm
Female	0.246 n = 4936	0.262 n = 4828	0.284 n = 4921	0.277 n = 4906	0.266 n = 4886
Male	0.222 n = 2887	0.253 n = 3012	0.274 n = 2888	0.267 n = 2891	0.263 n = 2974

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and sex.

4.2.2 Primary outcome regression with sex as moderator

We tested whether males and females differently responded to risk messages in an OLS regression. We performed one regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms).

Results are reported in Web Appendix Table 25. There was not a significant difference in vaccination by sex, and there were no significant interactions between study arm and sex.

Web Appendix Table 25. Study 3 OLS regression results for vaccination by study arm and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.276	0.269 – 0.283	76.009	39123	<0.001
Study arm: Passive control vs. high-risk messages	-0.029	-0.044 – -0.015	-4.060	39123	<0.001
Study arm: Active control vs. high-risk messages	-0.013	-0.028 – 0.001	-1.824	39123	0.068
Sex	-0.008	-0.019 – 0.004	-1.300	39123	0.194
Study arm: Passive control vs. high risk * Sex	-0.016	-0.040 – 0.007	-1.366	39123	0.172
Study arm: Active control vs. high risk * Sex	-0.001	-0.025 – 0.022	-0.117	39123	0.907

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Sex is a binary variable coded as 1 for males and 0 for females.

4.3 Age

4.3.1 Age 65+ vs. under 65

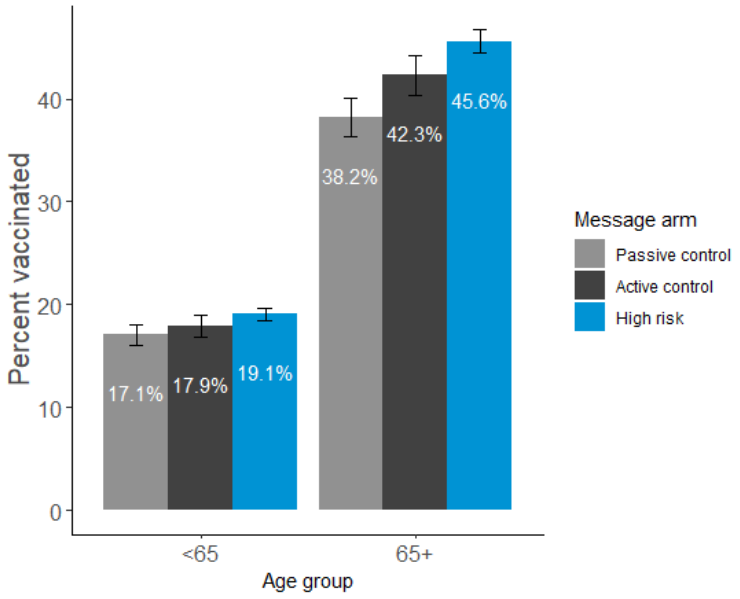
As in Studies 1 and 2, we tested whether vaccination differed in patients age 65+ compared with patients under age 65. We ran one OLS regression that included a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms). We crossed this study arm variable with age group (65+ vs. under 65).

OLS regression results are shown in Web Appendix Table 26 and Web Appendix Fig. 5. Patients ages 65+ (n = 12,299) were 26.5 percentage points more likely to get vaccinated than patients under age 65 (n = 26,830) in the high-risk message arms. We also found a significant interaction between age group and message arm, such that high-risk messages, relative to passive control, were significantly more effective in patients ages 65+ than those under age 65. The same interaction with active control was not significant. Overall, this pattern of results suggests that it may be particularly effective to inform older patients in the top 11–20% of risk that they are at high risk.

Web Appendix Table 26. Study 3 OLS regression results for vaccination by study arm and age group (65+ vs. under 65).

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.191	0.184 – 0.197	57.211	39123	<0.001
Study arm: Passive control vs. high-risk messages	-0.020	-0.033 – -0.007	-3.013	39123	0.003
Study arm: Active control vs. high-risk messages	-0.011	-0.025 – 0.002	-1.687	39123	0.092
Age group: 65+	0.265	0.253 – 0.277	44.217	39123	<0.001
Study arm: Passive control vs. high-risk messages * Age 65+	-0.053	-0.077 – -0.030	-4.463	39123	<0.001
Study arm: Active control vs. high-risk messages * Age 65+	-0.021	-0.044 – 0.002	-1.790	39123	0.073

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Age is a binary variable coded as 1 for patients ages 65+ and 0 for patients under age 65.



Web Appendix Fig. 5. Vaccination rate as a function of age group and risk messages in Study 3. The primary outcome plotted as a function of study arm and age group. Error bars indicate 95% confidence intervals.

4.3.2 Binned age

As in Studies 1 and 2, we examined whether response to messages differed as a function of age group. Sample sizes for each age bin were as follows: 18–24 *n* = 3,379, 25–34 *n* = 5,279, 35–44 *n* = 5,536, 45–54 *n* = 5,754, 55–64 *n* = 6,882, 65+ *n* = 12,299

To test whether age bin moderated study arm effects, we ran one OLS regression testing for an interaction between age bin and a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. This approach allowed for direct comparisons between high-risk message arms and both active control and passive control in the same regression (but note that high-risk message arms are coded as the baseline group, so negative values indicate *higher* vaccination in high-risk message arms). For the age bin variable, the Age 18–24 bin was the reference group.

Results are reported in Web Appendix Table 27. Vaccination rates increase with increasing age. Results are consistent with those reported in Section 9.3.4.1, where we reported a significant interaction between age group and study arm. The intervention effect (high risk vs. passive control) was significantly larger in the 65+ age bin compared with the reference group of age 18–24. There was also a larger intervention effect (high risk vs. passive control) for patients ages 45–54 compared with the age 18–24 age bin. Generally, for patients at moderately high risk, high-risk messages appear to be more effective with older age.

Web Appendix Table 27. Study 2 OLS regression results for vaccination by study arm and binned age.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.093	0.075 – 0.111	9.918	39111	<0.001
Study arm: Passive control vs. high-risk messages	-0.002	-0.039 – 0.034	-0.119	39111	0.906
Study arm: Active control vs. high-risk messages	0.008	-0.028 – 0.045	0.455	39111	0.649
Age 25-34	0.029	0.006 – 0.053	2.466	39111	0.014
Age 35-44	0.067	0.044 – 0.090	5.645	39111	<0.001
Age 45-54	0.114	0.091 – 0.137	9.721	39111	<0.001
Age 55-64	0.210	0.187 – 0.232	18.381	39111	<0.001
Age 65+	0.363	0.342 – 0.384	34.299	39111	<0.001
Study arm: Passive control vs. high-risk messages * Age 25-34	0.001	-0.045 – 0.048	0.056	39111	0.955
Study arm: Active control vs. high-risk messages * Age 25-34	-0.014	-0.061 – 0.033	-0.585	39111	0.559
Study arm: Passive control vs. high-risk messages * Age 35-44	-0.023	-0.069 – 0.023	-0.968	39111	0.333
Study arm: Active control vs. high-risk messages * Age 35-44	-0.026	-0.073 – 0.020	-1.113	39111	0.266
Study arm: Passive control vs. high-risk messages * Age 45-54	-0.046	-0.092 – -0.000	-1.977	39111	0.048
Study arm: Active control vs. high-risk messages * Age 45-54	-0.027	-0.073 – 0.019	-1.146	39111	0.252
Study arm: Passive control vs. high-risk messages * Age 55-64	-0.011	-0.056 – 0.034	-0.488	39111	0.625
Study arm: Active control vs. high-risk messages * Age 55-64	-0.025	-0.070 – 0.019	-1.107	39111	0.268
Study arm: Passive control vs. high-risk messages * Age 65+	-0.071	-0.112 – -0.030	-3.390	39111	0.001
Study arm: Active control vs. high-risk messages * Age 65+	-0.041	-0.082 – 0.000	-1.956	39111	0.051

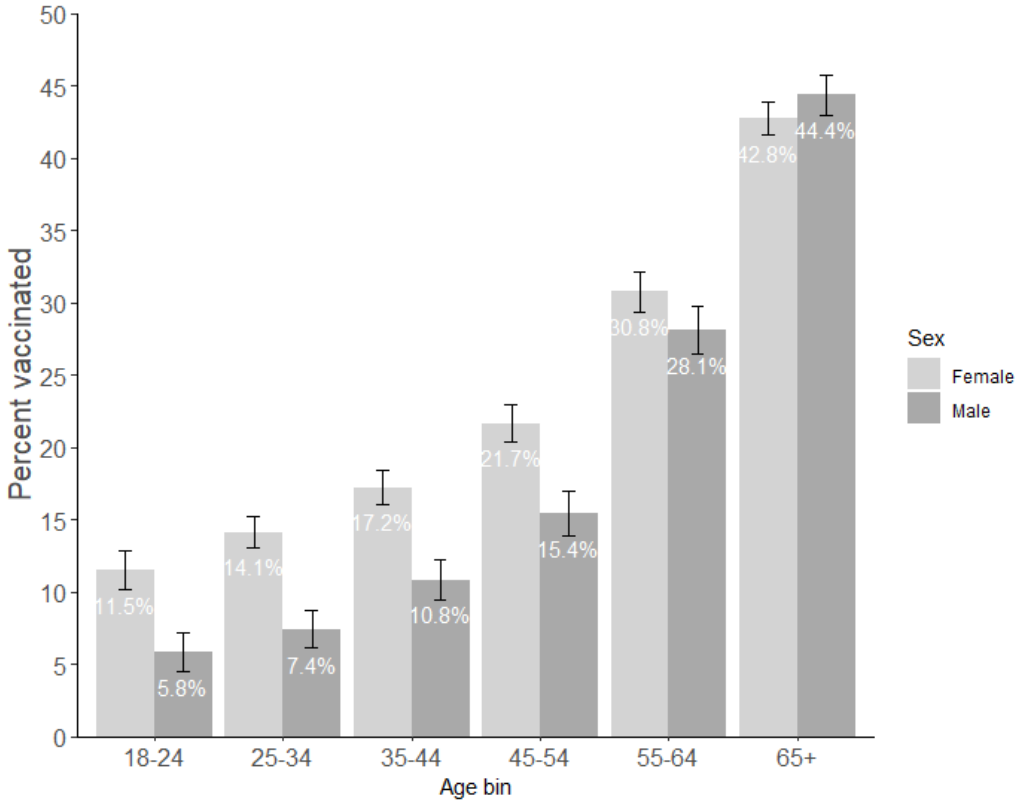
Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. The reference age group is age 18-24.

4.4 Age and sex

We tested for effects of study arm with sex, age bins, and interactions between sex and age as covariates in a preregistered exploratory analysis. As in the above exploratory analyses of Study 2 data, we ran one OLS regression testing for an interaction between age bin and a categorical variable to indicate study arm with any high-risk arm coded as 0, passive control coded as 1, and active control coded as 2. We crossed this variable with age bin.

With the age, sex, and interaction covariates included in the regression, there was a significant intervention effect of high-risk messages compared with both active and passive control messages. Both effects were stronger than the effects reported in the main text without covariates (Web Appendix Table 28).

Similar to the pattern in Studies 1 and 2, we found significant interactive effects between age group and sex, such that the vaccination pattern by sex was significantly different in the 65+ age group than the 18–24 age group. Generally, females appear to be more likely to be vaccinated at younger ages, while males are to be increasingly likely to be vaccinated (relative to females) with increasing age (Web Appendix Fig. 6, Web Appendix Table 28).



Web Appendix Fig. 6. Vaccination rate as a function of sex and age bin in Study 3.

The percentage of male and female patients in each age bin who were vaccinated during the primary outcome time frame. Error bars indicate 95% confidence intervals.

Web Appendix Table 28. Study 3 OLS regression results for vaccination by study arm, age bin, and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.126	0.108 – 0.144	13.601	39115	<0.001
Study arm: Passive control vs. high-risk messages	-0.036	-0.047 – -0.026	-6.661	39115	<0.001
Study arm: Active control vs. high-risk messages	-0.018	-0.028 – -0.007	-3.213	39115	0.001
Age 25-34	0.026	0.004 – 0.049	2.308	39115	0.021
Age 35-44	0.057	0.035 – 0.079	4.998	39115	<0.001
Age 45-54	0.101	0.079 – 0.124	8.873	39115	<0.001
Age 55-64	0.192	0.170 – 0.214	17.121	39115	<0.001
Age 65+	0.313	0.292 – 0.333	30.227	39115	<0.001
Sex	-0.056	-0.086 – -0.027	-3.766	39115	<0.001
Age 25-34 * Sex	-0.012	-0.050 – 0.026	-0.606	39115	0.544
Age 35-44 * Sex	-0.007	-0.045 – 0.031	-0.363	39115	0.716
Age 45-54 * Sex	-0.006	-0.043 – 0.031	-0.326	39115	0.744
Age 55-64 * Sex	0.030	-0.006 – 0.065	1.631	39115	0.103
Age 65+ * Sex	0.072	0.039 – 0.105	4.298	39115	<0.001

Note. Study arm is coded as 0 for patients in any high-risk arm (high risk only, high risk medical records, high risk algorithm) 1 for patients in the passive control arm, and 2 for patients in the active control arm. Sex is coded as 1 for male and 0 for female. The reference age group is age 18-24.

4.4.2 Age, sex, and study arm

In Study 3, we also preregistered a 3-way age bin by sex by study arm interaction. The 3-way interaction term did not reach significance (ANOVA: $F(10, 39093) = 0.603$, $p = .813$).

5 Study 4 additional analyses

5.1 Race and ethnicity

5.1.1 Primary outcome by study arm, race, and ethnicity

Web Appendix Table 29. Vaccination rates by study arm and race in the primary analysis sample for Study 4.

Race	Passive control	SMS	Portal	Portal + SMS	Letter + portal + SMS	Letter
American Indian or Alaska Native	0.250 n = 12	0.100 n = 10	0.389 n = 18	0.071 n = 14	0.222 n = 18	0.125 n = 8
Asian	0.200 n = 10	0.250 n = 24	0.176 n = 17	0.105 n = 19	0.250 n = 12	0.200 n = 10
Native Hawaiian or Other Pacific Islander	0.250 n = 16	0.222 n = 9	0.143 n = 14	0.071 n = 14	0.250 n = 12	0.100 n = 20
Black or African American	0.148 n = 169	0.090 n = 188	0.085 n = 165	0.119 n = 176	0.141 n = 184	0.097 n = 176
White	0.183 n = 4607	0.189 n = 4601	0.194 n = 4620	0.201 n = 4603	0.200 n = 4611	0.212 n = 4624
More Than One Race	0.500 n = 2	0 n = 0	0 n = 0	0 n = 1	0 n = 0	0 n = 0
Unknown	0.103 n = 39	0.121 n = 33	0.038 n = 26	0.139 n = 36	0.212 n = 33	0.086 n = 35

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and racial group. Results i

Web Appendix Table 30. Vaccination rates by study arm and ethnicity in the primary analysis sample for Study 4.

Ethnicity	Passive control	SMS	Portal	Portal + SMS	Letter + portal + SMS	Letter
Hispanic or Latino	0.128 n = 226	0.137 n = 234	0.109 n = 247	0.123 n = 228	0.149 n = 222	0.141 n = 248
Not Hispanic or Latino	0.184 n = 4595	0.188 n = 4597	0.195 n = 4585	0.200 n = 4599	0.200 n = 4623	0.211 n = 4592
Unknown	0.176 n = 34	0.147 n = 34	0.143 n = 28	0.250 n = 36	0.240 n = 25	0.091 n = 33

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and ethnic group.

5.1.2 Primary outcome regression with race/ethnicity as moderator

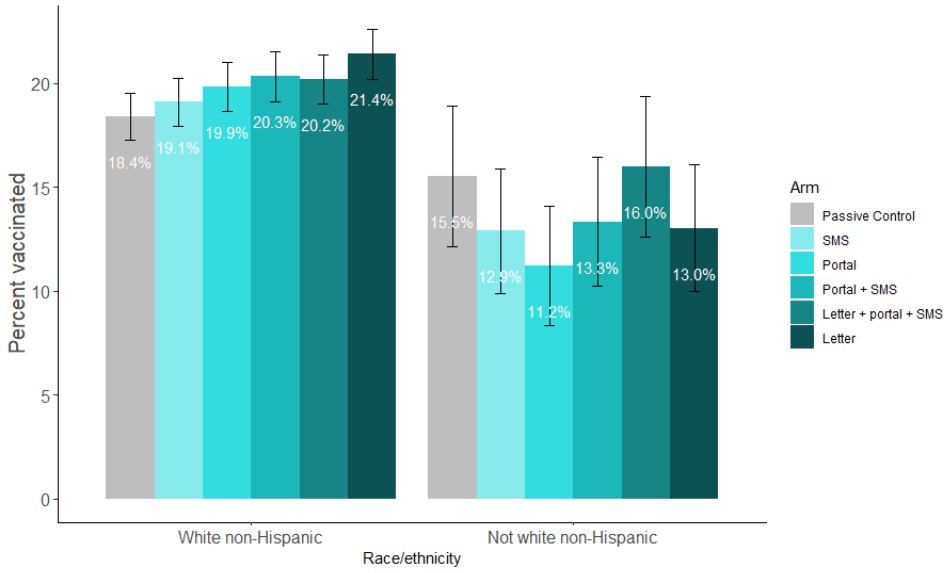
We tested whether modalities were differently effective as a function of race/ethnicity. As in Studies 1–3, the combined race/ethnicity predictor variable was set to 1 for patients who were both white and non-Hispanic and 0 for patients who were not both white and non-Hispanic (or for whom race or ethnicity were unknown). Among patients in the primary analysis sample, 26,451 patients were white and non-Hispanic, and 2,735 patients were not white and non-Hispanic.

Results are reported in Web Appendix Table 31 and Web Appendix Fig. 7. Both portal messages and letters, compared with no messages, were differentially associated with significantly more vaccinations among white non-Hispanic patients than among those who were not.

Web Appendix Table 31. Study 4 regression results for vaccination by study arm and race/ethnicity.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.155	0.118 – 0.192	8.243	29174	<0.001
Study arm: SMS	-0.026	-0.078 – 0.025	-1.006	29174	0.315
Study arm: Portal	-0.043	-0.094 – 0.009	-1.634	29174	0.102
Study arm: Portal + SMS	-0.022	-0.073 – 0.030	-0.826	29174	0.409
Study arm: Letter + portal + SMS	0.005	-0.047 – 0.057	0.175	29174	0.861
Study arm: Letter	-0.025	-0.076 – 0.027	-0.943	29174	0.346
Race/ethnicity	0.029	-0.010 – 0.068	1.459	29174	0.145
Study arm: SMS * Race/ethnicity	0.033	-0.021 – 0.087	1.211	29174	0.226
Study arm: Portal * Race/ethnicity	0.057	0.003 – 0.111	2.082	29174	0.037
Study arm: Portal + SMS * Race/ethnicity	0.041	-0.013 – 0.095	1.485	29174	0.138
Study arm: Letter + portal + SMS * Race/ethnicity	0.013	-0.041 – 0.068	0.477	29174	0.634
Study arm: Letter * Race/ethnicity	0.055	0.001 – 0.109	1.987	29174	0.047

Note. The reference group for the study arm variable is passive control. Race/ethnicity is a binary variable coded as 1 for patients who were both white and non-Hispanic and 0 for those who were not white and non-Hispanic, for whom race/ethnicity were unknown.



Web Appendix Fig. 7. Vaccination rate as a function of race/ethnicity and study arm in the primary analysis sample in Study 4.

The primary outcome plotted as a function of study arm and patient race/ethnicity. Error bars represent 95% confidence intervals.

5.2 Sex

5.2.1 Primary outcome by study arm and sex

Web Appendix Table 32. Vaccination rates by study arm and sex in the primary analysis sample for Study 4.

Sex	Passive control	SMS	Portal	Portal + SMS	Letter + portal + SMS	Letter
Female	0.167 n = 3482	0.173 n = 3436	0.175 n = 3421	0.183 n = 3446	0.185 n = 3397	0.191 n = 3434
Male	0.218 n = 1373	0.213 n = 1429	0.227 n = 1439	0.231 n = 1417	0.229 n = 1473	0.243 n = 1439

Note. Within each cell, the top number is the proportion of patients who were vaccinated during the primary outcome time frame. n reflects the number of patients in the indicated arm and sex.

5.2.2 Primary outcome regression with sex as moderator

Within the primary analysis sample, we tested whether males and females differently responded to modalities or modality combinations in an OLS regression. We ran one regression that included a categorical variable to indicate study arm with passive control as the baseline group.

Results are reported in Web Appendix Table 33. Sex does not significantly moderate intervention effects for any study arms relative to passive control.

Web Appendix Table 33. Study 4 OLS regression results for vaccination by study arm and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.167	0.154 – 0.180	24.987	29174	<0.001
Study arm: SMS	0.006	-0.012 – 0.025	0.666	29174	0.506
Study arm: Portal	0.008	-0.010 – 0.027	0.868	29174	0.385
Study arm: Portal + SMS	0.016	-0.003 – 0.035	1.686	29174	0.092
Study arm: Letter + portal + SMS	0.018	-0.001 – 0.036	1.864	29174	0.062
Study arm: Letter	0.024	0.006 – 0.043	2.551	29174	0.011
Sex	0.052	0.027 – 0.076	4.113	29174	<0.001
Study arm: SMS * Sex	-0.011	-0.046 – 0.023	-0.644	29174	0.519
Study arm: Portal * Sex	-0.000	-0.035 – 0.034	-0.011	29174	0.991
Study arm: Portal + SMS * Sex	-0.004	-0.038 – 0.031	-0.209	29174	0.834
Study arm: Letter + portal + SMS * Sex	-0.007	-0.041 – 0.028	-0.384	29174	0.701
Study arm: Letter * Sex	-0.000	-0.035 – 0.034	-0.008	29174	0.994

Note. The reference group for the study arm variable is passive control. Sex is a binary variable coded as 1 for males and 0 for females.

5.3 Age

As with Studies 1–3, we examined whether response to messages differed as a function of age group within the primary analysis sample. Sample sizes for each age bin were as follows: 18–24 *n* = 2,005, 25–34 *n* = 4,242, 35–44 *n* = 4,655, 45–54 *n* = 4,937, 55–64 *n* = 5,442, 65+ *n* = 7,905.

To test whether age bin moderated study arm effects, we ran one OLS regression testing for an interaction between age bin and a categorical variable to indicate study arm. For the age bin variable, the Age 18–24 bin was the reference group.

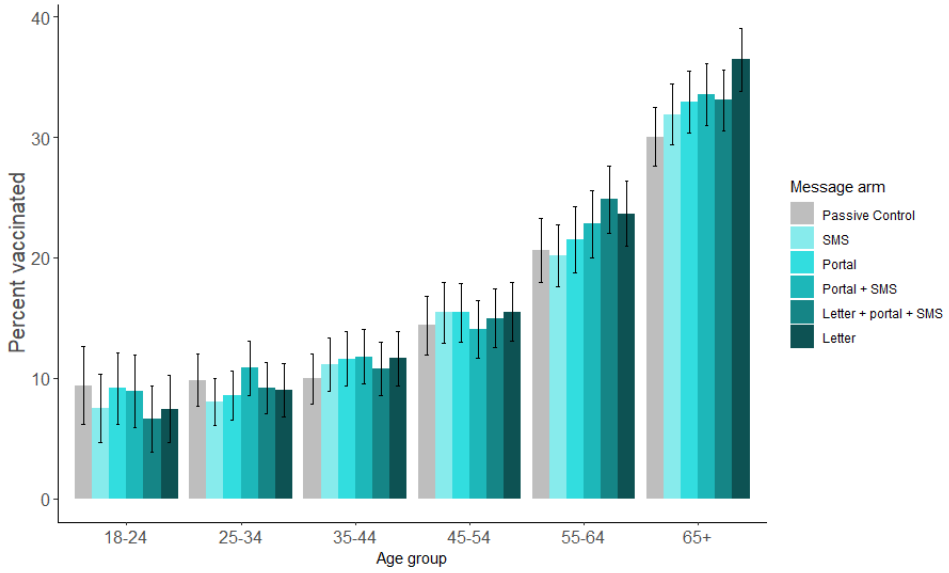
Results are reported in Web Appendix Table 34 and Web Appendix Fig. 8. Patients age 65+ compared with those ages 18–24 were significantly more likely to be vaccinated after being sent a letter (vs. passive control). Those ages 55–64, compared with those ages 18–24 were vaccinated at higher rates when sent a message via all three study modalities (vs. passive control).

Web Appendix Table 34. Study 4 OLS regression results for vaccination by age bin.

Predictors	Estimates	CI	*t*	*df*	*p*
(Intercept)	0.094	0.051 – 0.137	4.305	29150	<0.001
Study arm: SMS	-0.019	-0.078 – 0.040	-0.628	29150	0.530
Study arm: Portal	-0.002	-0.060 – 0.056	-0.074	29150	0.941
Study arm: Portal + SMS	-0.005	-0.063 – 0.054	-0.159	29150	0.873
Study arm: Letter + portal + SMS	-0.028	-0.088 – 0.032	-0.909	29150	0.364
Study arm: Letter	-0.019	-0.079 – 0.040	-0.644	29150	0.520
Age 25-34	0.004	-0.047 – 0.055	0.166	29150	0.868
Age 35-44	0.006	-0.045 – 0.056	0.214	29150	0.830
Age 45-54	0.050	-0.000 – 0.100	1.948	29150	0.051
Age 55-64	0.112	0.063 – 0.162	4.445	29150	<0.001
Age 65+	0.206	0.159 – 0.254	8.539	29150	<0.001
Study arm: SMS * Age 25-34	0.001	-0.070 – 0.072	0.027	29150	0.978
Study arm: Portal * Age 25-34	-0.010	-0.081 – 0.060	-0.282	29150	0.778
Study arm: Portal + SMS * Age 25-34	0.015	-0.056 – 0.086	0.413	29150	0.680
Study arm: Letter + portal + SMS * Age 25-34	0.022	-0.051 – 0.094	0.587	29150	0.557
Study arm: Letter * Age 25-34	0.011	-0.060 – 0.083	0.310	29150	0.757
Study arm: SMS * Age 35-44	0.031	-0.040 – 0.101	0.859	29150	0.390
Study arm: Portal * Age 35-44	0.019	-0.050 – 0.088	0.536	29150	0.592
Study arm: Portal + SMS * Age 35-44	0.023	-0.047 – 0.093	0.646	29150	0.518

Study arm: Letter + portal + SMS * Age 35-44	0.036	-0.035 – 0.108	1.005	29150	0.315
Study arm: Letter * Age 35-44	0.037	-0.034 – 0.107	1.017	29150	0.309
Study arm: SMS * Age 45-54	0.030	-0.041 – 0.100	0.829	29150	0.407
Study arm: Portal * Age 45-54	0.013	-0.056 – 0.082	0.368	29150	0.713
Study arm: Portal + SMS * Age 45-54	0.001	-0.068 – 0.071	0.041	29150	0.967
Study arm: Letter + portal + SMS * Age 45-54	0.034	-0.037 – 0.104	0.940	29150	0.347
Study arm: Letter * Age 45-54	0.031	-0.039 – 0.100	0.867	29150	0.386
Study arm: SMS * Age 55-64	0.014	-0.055 – 0.083	0.411	29150	0.681
Study arm: Portal * Age 55-64	0.011	-0.057 – 0.079	0.313	29150	0.754
Study arm: Portal + SMS * Age 55-64	0.026	-0.042 – 0.095	0.756	29150	0.450
Study arm: Letter + portal + SMS * Age 55-64	0.070	0.000 – 0.139	1.970	29150	0.049
Study arm: Letter * Age 55-64	0.050	-0.019 – 0.118	1.416	29150	0.157
Study arm: SMS * Age 65+	0.038	-0.028 – 0.104	1.121	29150	0.262
Study arm: Portal * Age 65+	0.031	-0.034 – 0.096	0.926	29150	0.354
Study arm: Portal + SMS * Age 65+	0.040	-0.025 – 0.106	1.199	29150	0.231
Study arm: Letter + portal + SMS * Age 65+	0.058	-0.008 – 0.125	1.716	29150	0.086
Study arm: Letter * Age 65+	0.084	0.018 – 0.150	2.480	29150	0.013

Note. The reference group for study arm is passive control. The reference age group is age 18-24.



Web Appendix Fig. 8. Vaccination rate as a function of age bin and message arm in Study 4.

Vaccination patterns in the primary outcome time frame by age bin and arm in the primary analysis sample. Error bars represent 95% confidence intervals.

5.4 Age and sex

Among those in the primary analysis sample, we tested for effects of study arm with sex, age bins, and interactions between sex and age as covariates.

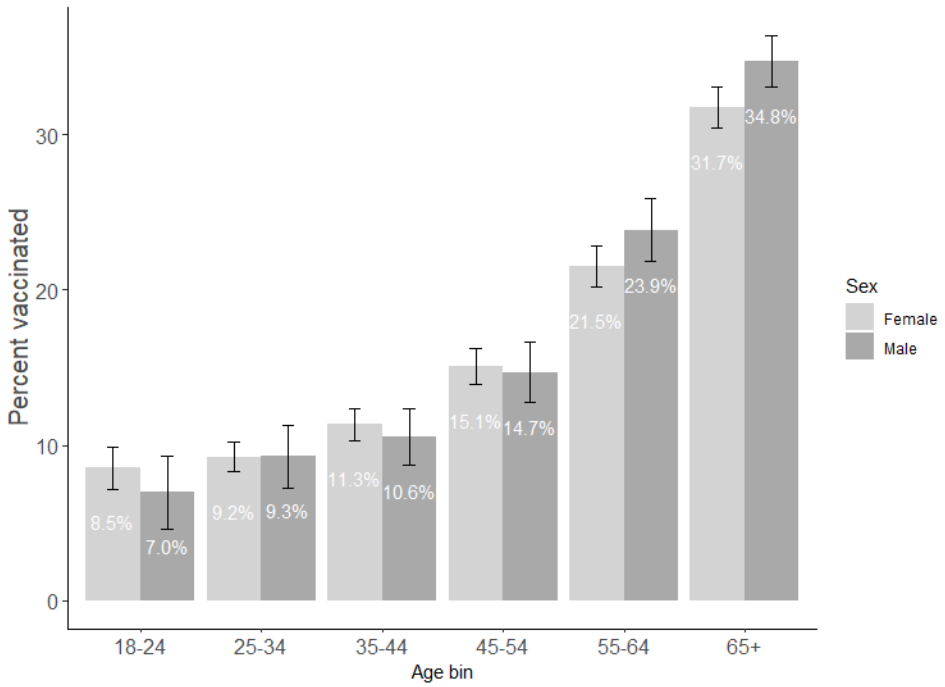
With the age, sex, and interaction covariates included in the regression, the effects of study arm did not substantially change, but vaccination in the portal + SMS arm was significantly higher than in the passive control arm (Web Appendix Table 35).

As in previous studies, relatively younger females were vaccinated at higher rates than males, but the pattern shifted with older age, such that males were increasingly likely to be vaccinated across age (Web Appendix Table 35, Web Appendix Fig. 9). Interestingly, in Study 4, this shift appears to have occurred at a slightly younger age, where male vaccination numerically surpassed female vaccination in the 55–64 age group. In contrast, for studies 1–3, vaccination was numerically higher in males only in the 65+ age group. This difference across studies may be due to differences in analysis samples across studies: the present Study 4 primary analysis sample (patients eligible for letter, portal, and SMS), is different demographically from patients eligible for fewer modalities.

Web Appendix Table 35. Study 4 OLS regression results for vaccination by study arm, age bin, and sex.

Predictors	Estimates	95% CI	*t*	*df*	*p*
(Intercept)	0.073	0.051 – 0.094	6.639	29169	<0.001
Study arm: SMS	0.004	-0.011 – 0.019	0.497	29169	0.619
Study arm: Portal	0.012	-0.003 – 0.027	1.563	29169	0.118
Study arm: Portal + SMS	0.017	0.002 – 0.032	2.215	29169	0.027
Study arm: Letter + portal + SMS	0.016	0.000 – 0.031	2.015	29169	0.044
Study arm: Letter	0.025	0.010 – 0.041	3.259	29169	0.001
Age 25-34	0.008	-0.015 – 0.030	0.647	29169	0.518
Age 35-44	0.028	0.006 – 0.051	2.434	29169	0.015
Age 45-54	0.065	0.043 – 0.088	5.637	29169	<0.001
Age 55-64	0.130	0.108 – 0.153	11.246	29169	<0.001
Age 65+	0.232	0.210 – 0.254	20.740	29169	<0.001
Sex	-0.016	-0.056 – 0.025	-0.764	29169	0.445
Age 25-34 * Sex	0.015	-0.035 – 0.066	0.606	29169	0.544
Age 35-44 * Sex	0.008	-0.040 – 0.055	0.307	29169	0.759
Age 45-54 * Sex	0.012	-0.035 – 0.059	0.505	29169	0.614
Age 55-64 * Sex	0.039	-0.007 – 0.085	1.656	29169	0.098
Age 65+ * Sex	0.046	0.002 – 0.090	2.050	29169	0.040

Note. The reference group for the study arm variable is passive control. Sex is a binary variable coded as 1 for males and 0 for females. The reference age group is age 18-24.



Web Appendix Fig. 9. Vaccination rate as a function of sex and age bin in Study 4.

The percentage of male and female patients in each age bin who were vaccinated during the primary outcome time frame. Error bars represent 95% confidence intervals.

5.4.1 Age, sex, and study arm

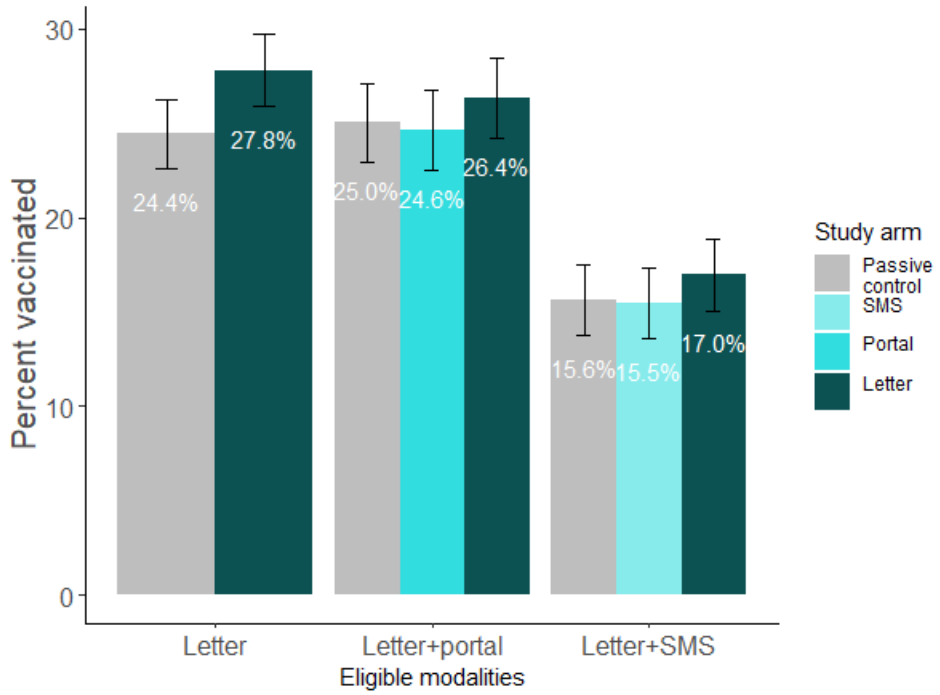
A preregistered regression including a 3-way age by sex by study arm interaction term did not explain significantly more variance in vaccination compared with a regression that included only 2-way interactions between age and sex, age and study arm, and sex and study arm ($F(25, 29114) = 0.780$, $p = .773$). This result suggests that the effectiveness of modalities and combinations of modalities did not vary as a function of both age and sex.

5.5 Results for different modality combinations

5.5.1 Full sample combined

Our statistical analysis plan included analyses across experimental arms with the full sample combined. However, because their baseline vaccination rates (Web Appendix Fig. 10) and demographic variables differed substantially across modality eligibility groups (Web Appendix Table 36), we decided that such an analysis would yield misleading results, so we do not presently report them. In the following sections, we report results of preregistered analyses

as a function of modality eligibility group. We excluded the few patients who were eligible only for SMS (*n* = 2) or portal and SMS (*n* = 8).



Web Appendix Fig. 10. Vaccination rates as a function of modality eligibility and study arm.

The percentage of patients in each study arm who were vaccinated during the primary outcome time frame by modalities for which they were eligible. Error bars represent 95% confidence intervals.

Web Appendix Table 36. Differences in baseline characteristics by modality eligibility.

	Eligible for letter only (n = 4292)	Eligible for letter and portal (n = 4906)	Eligible for letter and SMS (n = 4315)	Eligible for letter, portal and SMS (primary analysis sample; n = 29186)
Age				
Mean (SD)	71.2 (15.4)	66.0 (18.8)	55.9 (18.3)	51.8 (18.2)
Median [Min, Max]	74.0 [18.0, 104]	71.0 [18.0, 103]	59.0 [18.0, 100]	52.0 [18.0, 103]
Sex				
Female	2464 (57.4%)	3034 (61.8%)	2480 (57.5%)	20616 (70.6%)
Male	1828 (42.6%)	1872 (38.2%)	1835 (42.5%)	8570 (29.4%)
Race				
American Indian or Alaska Native	4 (0.1%)	9 (0.2%)	9 (0.2%)	80 (0.3%)
Asian	3 (0.1%)	4 (0.1%)	12 (0.3%)	92 (0.3%)
Native Hawaiian or Other Pacific Islander	5 (0.1%)	4 (0.1%)	25 (0.6%)	85 (0.3%)
Black or African American	73 (1.7%)	79 (1.6%)	327 (7.6%)	1058 (3.6%)
White	4193 (97.7%)	4785 (97.5%)	3904 (90.5%)	27666 (94.8%)
More Than One Race	0 (0%)	0 (0%)	3 (0.1%)	3 (0.0%)
Unknown	14 (0.3%)	25 (0.5%)	35 (0.8%)	202 (0.7%)
Ethnicity				
Hispanic or Latino	99 (2.3%)	87 (1.8%)	410 (9.5%)	1405 (4.8%)
Not Hispanic or Latino	4175 (97.3%)	4785 (97.5%)	3878 (89.9%)	27591 (94.5%)
Unknown	18 (0.4%)	34 (0.7%)	27 (0.6%)	190 (0.7%)

5.5.2 Letter only

Among patients who were only eligible for a letter, vaccination was 3.4 percentage points higher in the group that was sent a letter compared with the group that was not sent a letter (b = 0.034, [95% CI: 0.007,0.060], t(4290) = 2.508, p = .012).

5.5.3 Letter + portal

Among patients eligible for letter and patient portal (n = 4,906), neither a letter nor a portal message significantly increased vaccination relative to passive control (letter: b = 0.013, [95% CI: -0.017, 0.043], t(4903) = 0.868, p = .385; portal: b = 0.004, [95% CI: -0.026, 0.034], t(4903) = 0.261, p = .794). There was also not a significant difference in vaccination between those sent a letter and those sent a portal message (b = 0.017, [95% CI: -0.013, 0.047], t(3262) = 1.126, p = .260).

5.5.4 Letter + SMS

For patients eligible for letter and SMS (*n* = 4,315), vaccination did not significantly differ between those in the passive control group and those in the SMS group (*b* = 0.001, [95% CI: -0.025, 0.028], *t*(4312) = 0.098, *p* = .922) or those in the letter group (*b* = 0.013, [95% CI: -0.013, 0.040], *t*(4312) = 0.984, *p* = .325). There was also not a significant difference in vaccination between those sent a letter and those sent an SMS (*b* = 0.015, [95% CI: -0.012, 0.042], *t*(2878) = 1.078, *p* = .281).